Feedback

Workshop 10

Workshop 10 feedback

- In workshop 10 we looked at two different methods for solving the diffusion equation
- The spatial derivative was calculated with a finite difference expression in both cases but the time stepping was different
- Explicit time stepping: evaluate finite difference expression at the current time step
- Implicit time stepping: evaluate finite difference expression at the next time step
- The implicit time stepping method requires solving a system of linear equations which give rise to a sparse matrix equation

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3 Explicit time stepping

Question 1: How many time steps are required to compute the solution?

```
> ./diffusion_1D
Grid spacing       = 0.001
Time step          = 5e-07
End time           = 0.01
Calculation finished after 20000 time steps
```

20000 time steps
with the initial parameters

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3 Explicit time stepping

- Increase the number of grid points from 1000 to 2000 and re-run the program.

Question 2: How many time steps are required to compute the solution with the new spatial resolution and how is the time step scaling with Δx?

```
> ./diffusion_1D
Grid spacing        = 0.001
Time step           = 5e-07
End time            = 0.01
Calculation finished after 20000 time steps
```

Before

```
> ./diffusion_1D
Grid spacing        = 0.0005
Time step           = 1.25e-07
End time            = 0.01
Calculation finished after 80000 time steps
>
```

After

$\Delta x \rightarrow \dfrac{\Delta x}{2}$ and the number of time steps has increased 4x

$$\Delta t = \nu \frac{(\Delta x)^2}{K}$$

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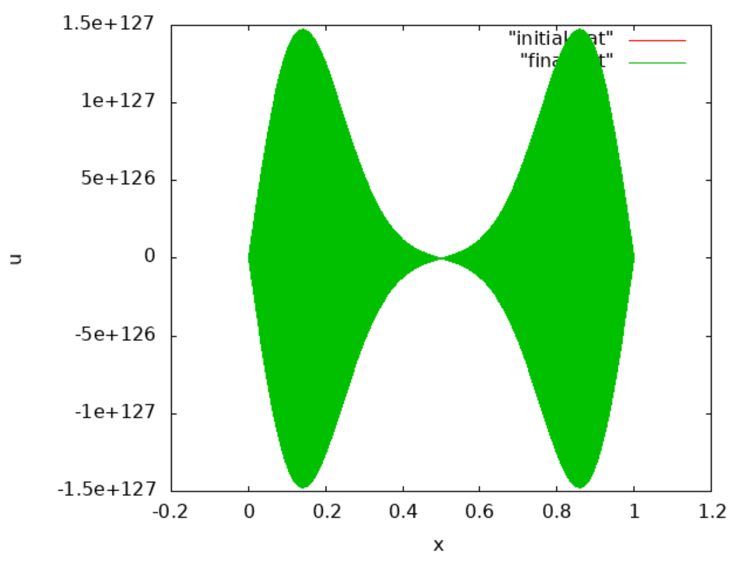
- We will now check the time step constraint by increasing the time step slightly. Set `nu=0.501` and re-run the program.

Question 3: Does the calculation produce valid results with the increased time step?

That will be a "no" …



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4 Implicit time stepping

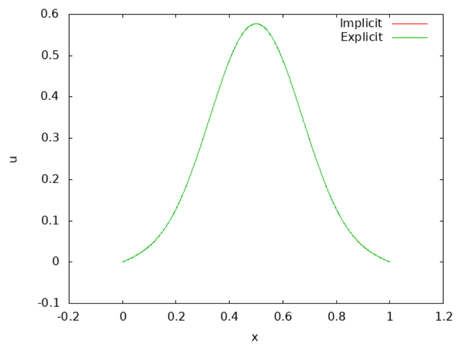
- Try increasing the time step by setting **nu=5.0** and re-run the program.

Question 4: How many time steps are now required and has this change noticeably affected the solution?

```
> emacs -nw diffusion_1D_im.c
> icc -mkl -std=c99 -o diffusion_1D_im diffusion_1D_im.c -lm
> ./diffusion_1D_im
Calling LAPACKE_dgttrf
Grid spacing      = 0.001
Time step         = 5e-06
End time          = 0.01
Calculation finished after 2000 time steps
>
```

2000 time steps

Curves are indistinguishable



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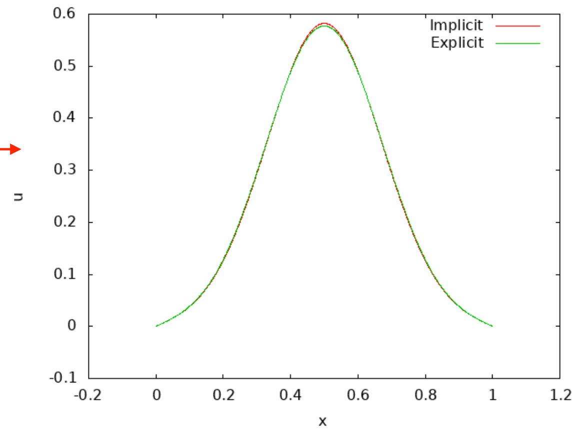
- Try increasing the time step even more by setting `nu=500.0`.

Question 5: How many time steps are now required and how has this change affected the solution?

```
> icc -mkl -std=c99 -o diffusion_1D_im diffusion_1D_im.c -lm
> ./diffusion_1D_im
Calling LAPACKE_dgttrf
Grid spacing       = 0.001
Time step          = 0.0005
End time           = 0.01
Calculation finished after 20 time steps
>
```

Some loss of accuracy but only 20 time steps required (was 20000 for the explicit solution)



Workshop 10 summary

- In this workshop we looked at solving the diffusion equation using explicit and implicit time stepping
- Explicit time stepping is subject to a stability constraint which scales as $(\Delta x)^2$
- Calculating the solution using implicit time stepping requires solving a sparse linear system
- However the implicit solution is unconditionally stable
- This means that we can take much larger time steps but will lose some accuracy